

RICHMOND, Australia

WMO: 943410

Lat: 20.701S

Lon: 143.120E

Elev: 679

StdP: 14.34

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.3	45.1	13.1	11.0	64.4	16.3	12.9	65.0	22.4	67.6	20.7	68.1	6.4	130	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	24.7	105.0	68.9	103.0	68.7	101.1	68.7	77.7	88.9	76.8	88.1	75.9	87.4	10.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
(p)	(q)	(r)	(s)	(t)	(u)	(v)	(w)	(x)	(y)	(z)	(aa)	(ab)	(ac)	(ad)	
75.3	136.0	81.1	74.1	130.6	80.7	73.1	126.3	80.5	41.8	88.8	40.8	88.4	40.0	86.8	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.2	20.1	17.9	35.8	108.8	2.3	2.5	34.1	110.6	32.8	112.1	31.5	113.5	29.9	115.3		
(5)			30.7	82.4	2.2	3.5	29.1	84.9	27.8	86.9	26.6	88.9	24.9	91.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	77.3	85.9	84.8	82.6	77.9	70.8	65.2	64.7	67.0	75.0	81.4	85.5	87.5 (6)
(7)		DBStd	9.61	4.31	4.02	3.81	4.36	5.46	6.22	5.79	5.77	5.51	5.30	4.27	4.14 (7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0 (8)
(9)		HDD65	220	0	0	0	1	15	76	78	46	3	1	0	0 (9)
(10)		CDD50	9962	1113	974	1010	837	644	455	455	526	749	972	1065	1162 (10)
(11)		CDD65	4708	648	554	545	388	194	81	68	108	302	508	615	697 (11)
(12)		CDH74	63449	8629	7206	6826	4639	2179	960	1013	1761	4134	7141	8837	10124 (12)
(13)		CDH80	35330	5025	4081	3745	2343	780	230	248	635	2163	4249	5475	6356 (13)
(14)	Wind	WSAvg	9.4	8.9	9.1	9.1	9.5	9.5	9.0	9.4	9.4	10.0	9.7	9.6	(14)
(15)	Precipitation	PrecAvg	19.3	5.0	4.1	2.3	0.9	0.6	0.4	0.3	0.2	0.2	0.7	1.2	3.0 (15)
(16)		PrecMax	41.5	26.1	11.1	12.0	8.1	2.7	4.2	3.0	2.8	2.3	5.2	8.0	9.8 (16)
(17)		PrecMin	7.9	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3 (17)
(18)		PrecStd	7.8	4.9	2.8	2.6	1.7	0.8	0.9	0.7	0.5	0.5	1.1	1.5	2.2 (18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	107.6	105.2	101.8	97.4	92.9	87.9	88.9	93.3	99.7	104.4	106.0	107.4	(19)
(20)		0.4% MCWB	71.8	69.9	69.6	68.1	66.5	64.1	64.5	62.9	64.9	66.6	68.4	68.7	(20)
(21)		2% DB	103.8	101.4	99.0	95.1	89.4	85.0	85.3	89.7	96.7	101.7	103.5	105.0	(21)
(22)		2% MCWB	70.8	70.8	69.8	67.2	65.3	63.5	62.5	61.6	63.7	66.0	67.6	68.9	(22)
(23)		5% DB	100.6	98.9	96.9	93.2	87.2	82.7	82.6	86.4	94.0	99.2	101.5	102.8	(23)
(24)		5% MCWB	71.3	70.9	69.3	66.6	64.0	62.2	60.5	60.4	63.0	65.2	67.3	68.9	(24)
(25)		10% DB	97.8	96.6	94.7	91.2	84.6	80.2	80.1	83.4	91.4	96.8	99.2	100.5	(25)
(26)		10% MCWB	71.4	71.1	69.6	66.4	63.3	60.8	59.2	59.0	62.1	64.9	67.1	69.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	79.2	78.8	78.3	74.9	71.0	67.7	68.5	67.3	70.4	72.5	75.4	77.4	(27)
(28)		0.4% MCDB	91.4	89.6	86.7	86.9	84.9	81.3	80.7	84.6	87.1	86.6	90.6	89.8	(28)
(29)		2% WB	77.8	77.4	76.8	72.7	68.9	65.9	65.7	64.9	68.4	70.2	73.6	76.1	(29)
(30)		2% MCDB	90.0	88.5	85.7	85.5	82.4	79.9	79.8	82.4	86.1	88.8	89.6	89.6	(30)
(31)		5% WB	76.7	76.6	75.6	71.0	67.1	64.2	63.5	62.8	66.5	68.8	72.3	75.0	(31)
(32)		5% MCDB	88.8	87.6	85.3	84.8	80.7	78.7	77.9	81.0	85.7	89.5	88.9	89.3	(32)
(33)		10% WB	75.7	75.7	74.1	69.3	65.5	62.7	61.2	60.8	64.4	67.4	71.1	74.0	(33)
(34)		10% MCDB	87.3	86.5	84.9	84.4	80.1	76.7	76.2	79.2	85.5	89.2	88.4	88.2	(34)
(35)	Mean Daily Temperature Range	MDBR	21.5	21.9	23.5	26.2	27.0	27.7	29.5	31.6	31.1	29.2	26.7	24.7	(35)
(36)		5% DB MCDBR	26.6	25.9	26.6	27.8	28.1	28.0	31.0	34.0	32.9	31.3	29.4	28.3	(36)
(37)		5% WB MCWBR	9.3	8.4	8.8	10.8	12.0	12.7	14.5	16.0	14.3	13.5	11.9	11.1	(37)
(38)		5% WB MCDBR	20.3	19.5	19.5	22.0	22.4	24.4	25.6	28.7	27.5	26.4	24.3	22.3	(38)
(39)	Clear-Sky Solar Irradiance	taub	0.385	0.381	0.360	0.345	0.329	0.316	0.308	0.324	0.363	0.385	0.391	0.404	(40)
(41)		taud	2.493	2.499	2.538	2.530	2.511	2.530	2.534	2.463	2.371	2.332	2.367	2.378	(41)
(42)		Ebn at Noon	305	302	300	292	285	283	290	295	295	297	300	298	(42)
(43)		Edh at Noon	37	36	33	31	30	28	28	33	39	42	42	42	(43)
(44)	All-Sky Solar Radiation	RadAvg	2224	2218	2070	1871	1557	1412	1546	1842	2154	2349	2394	2331	(44)
(45)		RadStd	193	132	163	140	82	104	72	77	123	136	126	166	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page